

MORUYA HEADS, Australia

WMO: 949370

Lat: 35.909S

Lon: 150.153E

Elev: 17

StdP: 101.12

Time Zone: 10.00 (AUS)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 4.4	(c) 5.3	(d) -1.6	(e) 3.3	(f) 13.8	(g) -0.4	(h) 3.7	(i) 12.8	(j) 14.8	(k) 13.1	(l) 13.6	(m) 12.9	(n) 2.9	(o) 250	(p) 0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 5.9	(c) 27.0	(d) 18.1	(e) 25.1	(f) 19.6	(g) 24.1	(h) 19.8	(i) 22.0	(j) 24.3	(k) 21.4	(l) 23.6	(m) 20.8	(n) 23.0	(o) 7.1	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	15.8	23.5	20.5	15.2	22.9	19.9	14.6	22.4	64.4	24.5	62.3	23.6	60.1	23.2	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 12.7	(c) 10.7	(e) 2.6	(f) 34.1	(g) 1.2	(h) 3.1	(i) 1.7	(j) 36.4	(k) 1.1	(l) 38.2	(m) 0.4	(n) 40.0	(o) -0.4	(p) 42.3		
(4) 13.9	(5) 12.7	(6) 10.7	(7) 2.6	(8) 34.1	(9) 1.2	(10) 3.1	(11) 1.7	(12) 36.4	(13) 1.1	(14) 38.2	(15) 0.4	(16) 40.0	(17) -0.4	(18) 42.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	19.8	20.2	19.1	16.6	14.6	12.6	11.4	12.1	13.9	15.2	16.7	18.4		
	DBStd	3.62	1.78	1.78	1.93	1.94	2.13	1.96	1.82	2.24	2.55	2.60	2.35	2.27		
	HDD10.0	11	0	0	0	0	0	1	7	3	0	0	0	0		
	HDD18.3	1110	6	3	12	59	118	172	216	194	137	106	61	27		
	CDD10.0	2144	303	285	283	197	143	79	49	68	116	161	201	259		
	CDD18.3	202	51	55	37	6	2	0	0	1	3	8	12	28		
	CDH23.3	479	86	91	73	14	3	0	0	3	24	71	58	57		
	CDH26.7	108	21	9	17	3	0	0	0	0	3	22	18	16		
(14) Wind	WSAvg	4.4	4.9	4.6	4.4	4.1	3.7	3.8	3.8	4.2	4.4	4.7	5.1	4.8		
(15) Precipitation	PrecAvg	1014	92	100	117	98	80	88	50	63	66	90	92	81		
	PrecMax	1755	362	576	571	506	260	514	224	310	197	525	302	247		
	PrecMin	468	10	2	6	2	1	0	0	1	3	7	9	10		
	PrecStd	320	70	121	108	110	65	101	53	73	49	95	67	61		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	26.8	27.1	25.1	23.0	20.7	19.6	23.4	27.0	30.1	29.0	28.5		
		MCWB	18.2	21.3	20.0	18.6	15.1	13.8	12.6	13.1	16.0	17.2	17.6	18.8		
	2%	DB	25.0	25.1	24.8	22.8	20.8	19.2	17.7	19.6	23.1	25.0	24.4	24.2		
		MCWB	20.3	21.5	20.3	18.1	15.6	13.5	11.6	12.2	14.4	16.0	17.4	20.2		
	5%	DB	23.9	24.3	23.6	21.7	19.7	17.9	16.5	18.0	20.5	21.3	21.7	23.0		
		MCWB	20.2	21.0	19.7	17.5	15.3	12.8	11.3	12.1	14.1	15.1	17.9	19.5		
	10%	DB	23.1	23.6	22.7	20.8	18.8	16.9	15.4	16.9	18.7	19.4	20.5	22.1		
		MCWB	19.7	20.5	19.0	16.9	14.8	12.2	10.9	12.1	14.0	15.1	17.2	18.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.0	22.3	20.2	18.5	16.0	14.2	15.9	17.7	18.6	20.2	21.7		
		MCDB	25.0	25.1	24.8	23.1	20.2	18.9	17.0	18.1	21.8	24.1	23.4	24.8		
	2%	WB	21.6	22.3	21.3	19.2	17.6	14.7	13.3	14.7	16.5	17.4	19.2	20.9		
		MCDB	23.9	24.4	23.6	21.7	19.7	18.0	15.7	17.5	20.1	21.1	22.5	23.5		
	5%	WB	21.0	21.6	20.4	18.3	16.6	13.8	12.4	13.7	15.6	16.6	18.3	20.0		
		MCDB	23.1	23.7	22.9	21.0	19.0	16.8	14.9	16.8	18.9	20.0	21.2	22.4		
	10%	WB	20.3	21.0	19.6	17.5	15.4	12.9	11.6	12.7	14.7	15.9	17.6	19.3		
		MCDB	22.5	23.1	22.2	20.1	17.8	15.8	14.4	16.0	17.8	19.0	20.0	21.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	5.9	6.6	7.7	7.6	8.1	8.4	9.1	8.7	8.0	6.8	6.4		
		MCDBR	7.4	6.9	8.1	8.9	8.5	8.9	9.5	10.2	11.8	12.3	9.5	7.9		
		MCWBR	4.2	4.2	4.7	5.5	5.3	5.2	5.6	5.7	5.4	5.4	4.6	4.7		
	5% WB	MCDBR	6.5	6.0	7.1	7.9	6.9	7.6	7.8	8.8	9.0	8.8	7.9	6.8		
		MCWBR	4.1	4.2	4.6	5.3	5.1	5.3	5.5	6.2	5.2	5.1	4.7	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.422	0.382	0.355	0.328	0.312	0.299	0.290	0.299	0.333	0.350	0.373	0.381		
	taud		2.349	2.504	2.561	2.610	2.601	2.633	2.654	2.622	2.521	2.504	2.475	2.470		
	Ebn at Noon		915	930	920	889	848	835	868	908	927	952	955	958		
	Edh at Noon		132	109	96	80	71	64	66	77	97	107	116	118		
(44) All-Sky Solar Radiation	RadAvg		6.49	5.60	4.78	3.72	2.88	2.23	2.63	3.49	4.75	5.70	6.14	6.66		
	RadStd		0.53	0.46	0.35	0.30	0.14	0.17	0.21	0.18	0.32	0.39	0.47	0.44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-75	+125	+52

Nomenclature: See separate page